



SCI_BUILD_A2_W02 · BUILD Science teacher note

Staff-facing; review only. Verbatim SoW objective: Compare and group rocks by their appearance and simple properties.

Source: BUILD scheme of work, Autumn 2 week 2. Pupil goal: I am learning to look at rocks and sort them.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can look and feel., I can sort two rocks., I can tell my sort rule.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	What are three ways to use less waste?	Use less, use again, recycle.	BUILD scheme of work, Autumn 2 week 1
2-4 weeks	How do animals get food?	From what they eat.	BUILD scheme of work, Autumn 1 week 7
last term	Which part helps you move with bones?	Muscles.	BUILD scheme of work, Autumn 1 week 4

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Point to the sample that feels rough. Answer: The sample with the rough surface; depend on the real kit. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
rock	Tier 3	a solid thing from the ground	I hold a safe rock.
grain	Tier 2	a small bit in a rock	I see tiny grains.
property	Tier 3	a thing you can check	This rock feels rough.
group	Tier 2	things put together by a rule	These rocks are smooth.

Faded example, tiered task and stretch

I do full: "This rock feels rough. I put it with the rough rocks." We do part-done: "This rock feels smooth. I put it with the rocks." Removed step: Name the group from an observed property. Independent: no filled step. Next fade: Show two unlabelled samples; pupil names both groups.

We do F: Feel one safe rock. Point to rough or smooth. M: Sort two real rocks. Say your rule. H: Can one rock fit two groups? Name each rule. Independent F: Look at two rocks. Point to one change. M: Sort two rocks by one thing you can check. H: Sort the same rocks in two ways. Tell why. Go deeper: Could one rock fit two groups? Show both rules. Named harder stretch — Rule test: Swap the rule. Which rocks move group, and why?



The kit route is hands-on or sensory: clean blunt rocks, optional hand lens, smooth and rough cards. Two on-screen interactions are card sort and Which step comes after look and feel?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Will the safe rocks feel the same? Kit: Two or more clean blunt rock samples, Optional hand lens, Practice cards. Steps: Look at each sample; Feel a safe face; Choose one sort rule; Place each rock in a group; Check each group against the rule. Results headings: Rock, Feature, Group. Check against the school risk assessment and CLEAPSS guidance. Check edges; no rock dust or tasting.

Hinge and decision

Question: Two rocks look alike. How can you sort them?

Option	Correct?	Diagnosis
Check one thing you can see or feel	yes	Uses an observed property.
Guess from the name	no	Uses a label instead of checking.
Call them the same	no	Misses a possible difference.
Use only a photo of colour	no	Assumes one cue tells all properties.

Rule: About 3 in 4 or more right → move on; otherwise re-model This rock feels rough. I put it with the rough rocks. Ask the parallel hinge. No scores or answers saved. Parallel: Two rocks look different. Can they both fit the smooth group? Feel them. Yes, if both are smooth.

Misconceptions:

Misconception	Spot	Correct
Rocks of the same colour must have the same feel.	Uses colour only, without touching.	Use safe samples to check rough or smooth as well.
A rock can belong to only one group.	Rejects the same rock from a new valid sort.	Show one sample can be smooth and dark; name the rule each time.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ? At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ? R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Give a three-step visual card for look closely at rocks; remove its last cue after a correct try.
Communication & Interaction	explicit instruction	Point to the real kit before a short spoken, signed or AAC response; wait for a turn.
SEMH	cognitive/metacognitive strategies	Offer a private first-then cue and planned pause before the hinge; re-enter with one step.
Sensory/Physical	technology	Use a stable tray, large cards or an accessible input; no forced touch.



Broad area	EEF five-a-day strategy	Specific support
EAL (language, not SEN)	flexible grouping	Rehearse a full M or H idea in any language before writing; keep the science challenge.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the object or model. → Ask: What will you check next? → Model one step and give the next turn back. → Finish one example together, then let the pupil try. No physical prompt. Stage scripts: I do “Watch my first step. What do you see?”; We do “What step is missing? Show it with the kit.”; Independent “What can you do first on your own?”. At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Point to a rough sample. → The checked rough rock in the kit; What is a rock? → A solid thing from the ground.

- M: How do you sort two rocks? → Check a feature such as rough or smooth; What is a property? → A thing you can check.

- H: Can one rock fit two groups? → Yes. It may be smooth and dark under two different rules; What does our diagram leave out? → Real rocks must be checked by sight or safe touch.

Regulation: Agree a quiet pause and supervised place. Give one short cue. Re-enter at the last completed step. No evidence while distressed.

lesson record: BUILD_SCI_A2_W02 via staff device or projector only. Gatsby 4: A geologist observes rock features before sorting samples.

Reading gate: BUILD below age 8; rough ARI grade 0.91 on 624 pupil-visible words (adult guidance and navigation excluded); heuristic age guide 6.91 years, not a formal reading-age test. This is a rough screen, not an individual assessment.

Pupil-facing risk: A rough edge may scratch skin.

Pupil-facing safety: Use clean blunt rocks. Keep them on the tray. Do not taste; wash hands after.

SLT planning and evaluation

Lesson code: SCI-BUILD-A2-W02

SoW reference: BUILD scheme of work, Autumn 2 week 2

Exact SoW outcome: Compare and group rocks by their appearance and simple properties.

Evidence (scheme of work): Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.



Label evidence on the school's evidence app with this code: SCI-BUILD-A2-W02. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Look at two safe rocks. Point to one you know. Staff: Offer a safe kit choice and short first-then cue.

Starter (3 min): Think back. Say, sign or point. Staff: Ask three low-stakes prior questions; answers in note.

I Do (5 min): Watch one rock sort. Staff: Pre-teach words; show the full model; ask the show-me check.

We Do (7 min): Feel two rocks. Sort safe samples. Staff: Hand over the missing step; offer F or H route. Reset once.

We Do 2 (5 min): Choose one answer. All of us show it. Staff: Take all pupil responses at once; use decision rule and parallel hinge.

Independent (8 min): Sort a safe rock. Show your rule. Staff: Choose one adult for supported pupils, one for group; swap across the week.

Review (4 min): Show one sort. Hear one next step. Staff: Give VF and one NS(+); invite one self-edit. Reset once.

Exit (3 min): Show your rule to an adult. Staff: Hear feedback and next step, choose next start, then one-minute reset.

Planned adaptation — cognition learning: Give a three-step visual card for look closely at rocks; remove its last cue after a correct try.

Planned adaptation — communication interaction: Point to the real kit before a short spoken, signed or AAC response; wait for a turn.

Planned adaptation — semh: Offer a private first-then cue and planned pause before the hinge; re-enter with one step.

Planned adaptation — sensory physical: Use a stable tray, large cards or an accessible input; no forced touch.

Planned adaptation — eal: Rehearse a full M or H idea in any language before writing; keep the science challenge.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: explicit instruction; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: flexible grouping

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Sort a safe rock. Show your rule. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W03. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W03: Test rocks for hardness and whether they let water through.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some



Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Name the group from an observed property. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "Two rocks look alike. How can you sort them?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Look at two safe rocks. Point to one you know. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Sort a safe rock. Show your rule. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use rock, grain, property, group with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.